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United States Patent	12415107
Kind Code	B1
Date of Patent	September 16, 2025
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Static structure with laser system for neutralizing embers

Abstract

A system for neutralizing embers is disclosed. The system includes a laser turret with a sensor array and a laser emitter configured to dispense laser energy. The laser turret can also include a control system configured to move the laser emitter. The laser control system can receive information from the sensor array and sending signals to the control system to move the laser emitter. Using these features, the laser control system can select an incoming ember based on a thermal signature associated with the incoming ember, and the laser control system can issue commands to the control system to direct the laser emitter towards the incoming ember, and the laser emitter can direct laser energy at the incoming ember. The system can also include provisions to track an ember and determine if an ember has been depleted of fuel.

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Appl. No.: 17/888750

Filed: August 16, 2022

Related U.S. Application Data

us-provisional-application US 63238811 20210831

Publication Classification

Int. Cl.: A62C3/02 (20060101); A62C37/38 (20060101); A62C99/00 (20100101)

U.S. Cl.:

CPC A62C37/38 (20130101); A62C3/0278 (20130101); A62C99/009 (20130101);

Field of Classification Search

CPC: A62C (37/38); A62C (37/40); A62C (31/03); A62C (3/0278); A62C (99/009); A62C (2/04); A62C (3/0214)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of Provisional Patent Application No. 63/238,811 filed Aug. 31, 2021, and titled “Static Structure with Laser System for Neutralizing Embers,” which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field of the Disclosure

(1) The present disclosure relates generally to a static structure with a laser system for neutralizing oncoming embers from a fire. In particular, the disclosure relates to a laser system using lasers to target and disrupt embers in the air around a static structure, building or dwelling.

2. Description of Related Art

(2) Destruction of buildings by wild fires is becoming more prevalent. The number of wild fires per year is increasing as people make greater use of developed and undeveloped areas of natural spaces. Both business, such as forestry, and personal uses, such as hunting and camping, contribute to the number of fires occurring in such outdoor spaces. With an increase in the number of users comes additional pressures upon the land, not only by increased fire risk, but also by increased numbers of less-well trained and inexperienced users. Increased pressure on use makes even a small fire a dangerous one.

(3) Global climate change also contributes to the danger of fires in outdoor spaces by drying the

fuel sources and strengthening storm winds, thus increasing the intensity of a fire. Wild fires tend to burn hotter, sending embers higher into the wind. Thus, embers are transported farther than they have been in the past, and are distributed throughout a larger area. As these hotter embers and smoke are carried further, they impinge upon more and more inhabited areas. Denser habitation means denser building, and more chance of building fires.

(4) Defense against wild fires is difficult. Many such fires burn in areas devoid of roads. Further, such natural spaces often are devoid of on-demand water sources, such as hydrants, and there often are but few people present to fight a fire. Many structures are uninhabited for months at a time.

Also, such fires may burn undetected, and so become well-established and difficult to extinguish.

(5) Further, the close proximity of buildings in urban and suburban regions increases the likelihood that a structure fire will spread from one building to another. Just as embers may spread a wild fire, embers may spread from one structure fire to another.

(6) There is a need in the art for a system and method that addresses the shortcomings of the prior art discussed above.

SUMMARY OF THE DISCLOSURE

(7) The disclosure is directed to use a system that disrupts embers in the air around a static structure. The system may detect embers, sweep or scan the lasers through the sky. To prevent the embers from entering the building, the lasers may be used to burn through all of the fuel in the ember while the ember is in the air. In some embodiments, the lasers may be focused on areas of building, such as vents, where embers may enter the building.

(8) In one aspect, the disclosure provides a static structure including a laser system comprising a laser turret. The laser turret may be disposed on the roof of the static structure, and can include a sensor array and a laser emitter configured to dispense laser energy. The laser turret can also include a control system configured to move the laser emitter. The laser control system can receive information from the sensor array and sending signals to the control system to move the laser emitter. Using these features, the laser control system can select an incoming ember based on a thermal signature associated with the incoming ember, and the laser control system can issue commands to the control system to direct the laser emitter towards the incoming ember, and the laser emitter can direct laser energy at the incoming ember.

(9) In another aspect, the laser control system uses a camera sensor to correct the direction of the laser emitter after the incoming ember has been initially acquired by the thermal signature associated with the incoming ember.

(10) In another aspect, the laser control system determines that a fuel level associated with the incoming ember has been depleted by sensing a discontinuity in the thermal signature of the incoming ember.

(11) In another aspect, the laser control system determines that a fuel level associated with the incoming ember has been depleted by sensing a rapid decline in the thermal signature of the incoming ember.

(12) In another aspect, the laser turret includes a rotating motor that rotates the laser turret with respect to the static structure.

(13) In another aspect, the laser control system tracks multiple incoming embers and calculates a velocity and trajectory of each of the incoming embers, the laser control system selecting at least one priority ember likely to contact the static structure for initial service, and wherein the laser control system directs laser energy at the priority ember.

(14) In another aspect, the laser control system continues to direct laser energy at the priority ember until a fuel level associated with the priority ember has been depleted.

(15) In another aspect, the disclosure provides a method for neutralizing burning embers that may burn a building, the method comprising the steps of scanning a region adjacent to the building to locate a first burning ember having a risk of burning the building; energizing a laser light emitter to emit laser light; and directing emitted laser light on the ember for a period sufficient to neutralize

the ember.

(16) Other systems, methods, features and advantages of the disclosure will be, or will become, apparent to one of ordinary skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features and advantages be included within this description and this summary, be within the scope of the disclosure, and be protected by the following claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure can be better understood with reference to the following drawings and description. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the disclosure. Moreover, in the figures, like reference numerals designate corresponding parts throughout the different views.
- (2) FIG. 1 is a schematic diagram of an embodiment of a static structure;
- (3) FIG. 2 is a schematic diagram of an embodiment of a roof turret of static structure;
- (4) FIG. 3 is a schematic diagram of an embodiment of side turrets of static structure;
- (5) FIG. 4 is a schematic flow diagram of a method for selecting and servicing a target, according to an embodiment;
- (6) FIG. 5 is a schematic diagram of a fire in close proximity to a static structure, in according to an embodiment;
- (7) FIG. 6 is a schematic diagram of an embodiment of a laser turret;
- (8) FIG. 7 is a schematic diagram of a laser turret tracking an incoming ember, according to an embodiment;
- (9) FIG. 8 is a schematic a laser turret directing energy towards an incoming ember, according to an embodiment;
- (10) FIG. 9 is a schematic a laser turret engaging an incoming ember, according to an embodiment;
- (11) FIG. 10 is a graph of the thermal signature of an ember over time, according to an embodiment;
- (12) FIG. 11 is a schematic diagram of a group of incoming embers and a laser turret, according to an embodiment;
- (13) FIG. 12 is a schematic diagram of a group of incoming embers and a laser turret, according to an embodiment; and
- (14) FIG. 13 is a schematic diagram of a group of incoming embers and a laser turret, according to an embodiment.

DETAILED DESCRIPTION

(15) Various embodiments provide a static structure or a system that includes provisions that help with fire prevention. Specifically, the various embodiments include features that help prevent alighted materials from settling on or near the static structure. This is accomplished by depleting the available fuel of the alighted material before it settles on the static structure.

(16) Referring to FIG. 1, static structure **100** is shown. Solely for purposes of illustration, static structure resembles a detached, single family dwelling. Static structure **100** could be any kind of static structure including a residential building or a commercial building. Examples of residential buildings include townhomes or row houses, apartment buildings, condominiums, multi-family dwellings, and detached single family homes. Examples of commercial buildings include office buildings, government buildings, retail developments, houses of worship, healthcare facilities, warehouses, storage facilities, and financial institutions. Principles and features of the present invention may be applied to any kind of static structure.

(17) As shown in FIG. 1, static structure **100** may include one or more laser turrets. In the

embodiment shown in FIG. 1, static structure **100** includes a roof turret **102** disposed centrally on roof **110** of static structure **100**. The example shown in FIG. 1 shows roof turret **102** being located centrally on roof **110**. FIG. 1 just shows an example, roof turret **102** may be placed in any suitable location on roof **110**, including peripheral portions or on the edge of roof **110**. Also, roof turret **102** is an optional feature. Some embodiments of static structure **100** may include a roof turret, while other embodiments of static structure **100** may exclude a roof turret. In some cases, static structure **100** may include multiple roof turrets.

(18) Some embodiments may optionally include first side turret **104**. First side turret **104** may be placed on a first side wall **112** of static structure **100**. As shown in FIG. 1, first side wall **112** is a structure that is generally vertically disposed and may be connected to roof **100**. Like roof turret **102**, first side turret **104** is an optional feature. Some embodiments of static structure **100** may include a first side turret, while other embodiments of static structure **100** may exclude a first side turret. In some cases, first side turret **104** may be disposed on a corner of first side wall **112**. Some embodiments may optionally include second side turret **106**. Second side turret **106** may be placed on a second side wall **114** of static structure **100**. As shown in FIG. 1, second side wall **114** is a structure that is generally vertically disposed and may be connected to roof **100**. In the embodiment shown in FIG. 1, second side wall **114** is distinct from first side wall **112**. Like roof turret **102**, second side turret **106** is an optional feature. Some embodiments of static structure **100** may include a second side turret, while other embodiments of static structure **100** may exclude a second side turret. Some embodiments may include a combination of two or more turrets selected from roof turrets and side turrets. Some embodiments of static structure **100** may include more than two side turrets.

(19) FIG. 2 is a schematic diagram of an embodiment of roof turret **102**. In this embodiment, roof turret **102** is centrally located on roof **110** and at a peak of roof **110**. This position provides roof turret **102** with a 360° arch sweep or arch coverage. From this position, roof turret **102** is capable of directing a laser beam to a target approaching from any direction.

(20) FIG. 3 is a schematic diagram of an embodiment of the placement of first side turret **104** and second side turret **106**. In this embodiment, first side turret **104** has an arch sweep or arch coverage of about 270°. First side turret **104** is unable to engage targets in defilade behind static structure **100**. This results in a 90° segment obscured by static structure **100**. Similarly, second side turret **106** also has an arch sweep or arch coverage of about 270°. Second side turret **106** is unable to engage targets in defilade behind static structure **100**. This results in a 90° segment obscured by static structure **100**. In the embodiment shown in FIG. 5, the two side turrets are disposed on opposite corners of static structure **100** so that the two side turrets working in combination will provide 360° coverage, with a couple of overlapping fields of fire.

(21) Some embodiments include provisions for directing a laser to a particular ember or incoming target. These embodiments can include the following optional steps of tracking one or more embers, selecting a particular ember, and servicing the selected ember. After this sequence has been completed, some embodiments may repeat the tracking, selecting and servicing sequence. An embodiment of a method for directing a laser at a particular ember is shown in FIG. 4.

(22) Referring to FIG. 4, the method can begin with step **402**, where a laser control system tracks multiple incoming embers. Referring to FIG. 5, a fire **502** close in proximity to static structure **100**, produces multiple embers **504**. Laser control system **550**, either disposed within laser turret **102** or in communication with laser turret **102**, may track incoming embers **504**. Referring to FIG. 6, which is a schematic diagram of an embodiment of a laser turret **102**, laser turret **102** may include a sensor array **604**. Sensor array **604** may optionally include a reflecting sensor **610**, an optical sensor **612**, and a thermal sensor **614**.

(23) Referring to FIGS. 4-6, laser control system **550**, using sensor array **604**, may track multiple incoming embers **504**. In some embodiments, laser control system **550** determines a predicted flight path of each ember by using the ember's current estimated velocity and direction. In the

embodiment shown in FIG. 4, first ember 510 is estimated to have insufficient velocity and trajectory to approach static structure 100. Likewise, second ember 512 is also estimated to have insufficient velocity and trajectory to contact or settle on static structure 100. Laser control system 550 calculates that fifth ember 518 has a velocity and trajectory that is likely to overshoot static structure 100 and laser control system 550 has determined that fifth ember 550 is unlikely to contact or settle on static structure 100.

(24) Unlike the other embers, third ember 514 and fourth ember 516 have been determined by laser control system 550 to present a velocity and trajectory profile that makes both third ember 514 and fourth ember 516 likely to either contact or settle on static structure 100. This determination by laser control system 550 of selecting particular embers that are likely to contact or settle on static structure 100 may form part of method step 404, where the laser control system 550 selects a particular ember. In some embodiments, this initial analysis of determining which embers are likely to contact static structure 100 may be used to distinguish embers that pose a risk to static structure 100 from embers that do not pose a risk to static structure 100.

(25) In addition, in some embodiments, laser control system 550 may prioritize embers that are likely to settle on the roof 110 of static structure 100 over embers that are likely to contact a side wall 112 of static structure 100. Embers that settle on roof 110 of static structure 100 may present an enhanced risk because those embers could have a longer opportunity, in terms of time, to ignite roof 110 of static structure 100. So laser control system 550 may use a second distinction between those embers that are likely to settle on the roof 110 versus those embers that may contact a side wall 112 of static structure 100. These distinctions may be used by laser control system 550 to prioritize and select particular embers for service.

(26) In the embodiment shown in FIG. 4, laser control system 550 has calculated, based on velocity and trajectory, that third ember 514 is likely to strike side wall 112 of static structure 100. Laser system 550 has also calculated that fourth ember 516 is likely to settle on roof 110 of static structure 100. Using just a calculated travel path for the embers, laser control system 550 may prioritize the fourth ember 516 for initial service. However, in some embodiments, laser system 550 may use additional factors to select the initial ember for service.

(27) Another factor that may be considered is an associated thermal signature of the ember. For example, embers with very high thermal signatures may indicate a relatively larger ember with relatively larger quantities of fuel available for combustion. Fuel being one of the elements of the fire triangle. The other elements of the fire triangle being heat and an oxidizer. In the context of the present invention, the oxidizer will most likely be ambient atmospheric oxygen. And the fuel will most likely be wood fuel. In some embodiments, laser control system 550 may include the thermal signature of an ember as a factor or consideration in determining priority. For example, in some cases, once the embers have been sorted between three categories: (1) likely to miss static structure 100, (2) likely to contact side wall of static structure, and (3) likely to settle on roof of static structure, then laser control system 550 can further prioritize those embers in the third group (likely to settle on roof) by thermal signature. In this example, laser control system 550 will select the “brightest” or “hottest” ember, the ember with the highest thermal signature, among the embers that are likely to settle on the roof of the static structure for priority service.

(28) FIG. 6 is a schematic diagram of an embodiment of laser turret 102. Laser turret 102 includes laser emitter 602 mounted on turret 606. Turret 606 may be mounted onto static structure 100, and include provisions that permit waist rotation so that turret 606 may rotate with respect to static structure 100. In the embodiment shown in FIG. 6, rotating motor 616 drives inner wall 618 of turret 606 to provide waist rotation of turret 606. In some embodiments, inner wall 618 and rotating motor 616 may include gears, teeth, splines or other engaging and cooperating surfaces. Some embodiments of turret 606 may include a bearing 619 that helps to facilitate waist rotation of turret 606 about static structure 100.

(29) Turret 606 may also include provisions that permit laser emitter 602 to elevate and depress

with respect to static structure **100**. In the embodiment shown in FIG. 6, turret **606** includes fulcrum **620** and powered actuator **622**. Powered actuator **622** is capable of pivoting laser emitter **602** about fulcrum **622**. This allows laser control system **550** to control the elevation and depression of laser emitter **602**. Laser turret **102** may also include provisions to detect and track incoming embers. Any desired sensor or device may be used to detect and track incoming embers. In the embodiment shown in FIG. 6, laser turret **102** includes sensor array **504**.

(30) As discussed above, Sensor array **604** may optionally include a reflecting sensor **610**, a camera sensor **612**, and a thermal sensor **614**. A reflecting sensor **610** is any type of sensor that emits an electromagnetic signal and detects a corresponding reflected signal. Examples of reflecting sensors include RADAR, SONAR, and LIDAR. The use of a reflecting sensor is optional. Camera sensors refer to a broad variety of sensors that are capable of detecting viable and non-visible light. Examples of camera sensors include Charged Coupled Devices (CCDs), Time of Flight sensors, CMOS image sensor, Structured Light Sensors, or any other sensor that can detect visible and non-visible light. Thermal sensor **614** is generally a sensor capable of detecting the relative temperature and emissivity of a distant object. Generally, an Infrared (IR) sensor may be used as a thermal sensor.

(31) Laser control system **550** may receive information from sensor array **604** and provide signals to various cooperating elements of turret **606** to control the operation of laser emitter **602**. In some embodiments, laser control system **550** may be disposed within turret **606**, as shown in FIG. 6. In this embodiment, sensor array **604** may communicate with laser control system **550** using physical wiring. Laser control system **550** may also communicate with rotating motor **616** and powered actuator **622** using physical wiring to direct laser emitter **602** in the desired direction. In other embodiments, laser control system **550** may be spaced from turret **606**. In these embodiments, laser control system **550** may receive information from sensor array **604** wirelessly. Laser control system **550** may also send control signals wirelessly to various elements, like rotating motor **616** and powered actuator **622**.

(32) FIGS. 7-9 show an embodiment of laser control system **550** targeting, tracking and servicing an incoming ember **416**. In FIG. 7, laser control system **550** has determined fourth ember **416** for priority service. In some embodiments, thermal sensor **614** may be used to initially target or aim laser emitter **502** towards fourth ember **416**. This may be advantageous because thermal sensor **614** may have been used to initially track and prioritize incoming targets. This persistent engagement with the potential target may allow thermal sensor **614** to provide an immediate firing solution to laser control system **550**. In other embodiments, reflecting sensor **610** may be used to initially target or aim laser emitter **502** towards fourth ember **416**. In some embodiments, both thermal sensor **614** and reflecting sensor **610** may be used in combination to initially target fourth ember **416**.

(33) In FIG. 8, laser control system **550** orders laser emitter **502** to emit laser beam **802** at fourth ember **416**. Camera sensor **612** may be used to determine if laser beam **802** has successfully contacted fourth ember **416**. In the embodiment shown in FIG. 8, laser beam **802** has missed fourth ember **416**. This miss may be captured by camera sensor **612**, and this information may be provided to laser control system **550**. In some embodiments, camera sensor **612** may be used to correct the aim of laser emitter **502**. In some cases, camera sensor **612** may continuously track and correct the aim of laser emitter **502**.

(34) Referring to FIG. 9, based on the received optical information from camera sensor **612**, laser control system **550** may correct the aim of laser emitter **502** until corrected laser beam **902** contacts fourth ember **416**. Camera sensor **612** may be used to continuously track fourth ember **416** during flight, and camera sensor **612** may be used to continuously correct the aim of laser emitter **502** so that corrected laser beam **902** remains in contact with fourth ember **416** as the fourth ember **416** continues along its trajectory.

(35) While corrected laser beam **902** remains in contact with fourth ember **416**, corrected laser

beam **902** services fourth ember **416** by irradiating fourth ember **416** with directed electromagnetic energy. This additional energy is used to accelerate the consumption of fuel contained within fourth ember **416**. In other words, laser beam **902** is intended to rapidly burn any remaining fuel in fourth ember **416**, thus depleting the fuel supply of fourth ember **416**. Ideally, the fourth ember **416** would be depleted of its fuel by corrected laser beam **902** prior to contacting static structure **100**. When fourth ember **416** has been depleted of its fuel, fourth ember **416** would likely generate far less heat and would be much less likely to ignite static structure **100** even if the depleted fourth ember settled on the roof **110** of static structure **100**.

(36) Some embodiments can include provisions for determining when an ember has been depleted and the next ember in order of priority may be targeted. Referring to FIG. **10**, which shows an embodiment of a graph of an ember's thermal signature over time, certain criteria may be used to shift from targeting one ember to another. In FIG. **10**, the X-axis is time in seconds and the Y-axis is the thermal signature of ember **416**. Ember **416** starts with an initial thermal signature **1002**. This thermal signature may have a slight decay pattern as shown in FIG. **10**. An embodiment of this condition is shown in FIG. **8**. A laser contacts fourth ember **416** at **1004**. An embodiment of this situation is shown in FIG. **9**. The laser may increase the heat of fourth ember **416** and increase its thermal signature to **1006**. The laser continues to contact the fourth ember **416** and burn its remaining fuel in region **1008**. At some point, the fuel of fourth ember **416** becomes depleted at **1010**. This causes a rapid decline in the thermal signature of fourth ember **416**, as shown in region **1012**.

(37) The depletion of fuel at **1010** causes a discontinuity of the thermal signature. In the embodiment shown in FIG. **10**, the laser burn phase in region **1008** produced a somewhat constant or slightly declining thermal signature. However, once the fuel has been depleted, as shown in region **1010**, a rapid decline in the thermal signature can be observed in region **1012**. If burn region **1008** and rapid decline region **1012** are modeled linearly, the slope of burn region **1008** would be different than the slope of rapid decline region **1012**. In one embodiment, the discontinuity between burn region **1008** and rapid decline region **1012** may be expressed as a difference in the slope of those regions, with the rapid decline region having a significantly more negative slope than the burn region. It can also be observed that the discontinuity occurs when the fuel has been depleted in region **1010**.

(38) In some embodiments, these observations are used to determine when a currently serviced ember has been depleted of fuel and the next ember, in terms of priority, may be targeted and serviced. In some cases, an observed discontinuity in the thermal signature of an ember may be used to infer or determine that the currently targeted ember has been depleted of fuel. In this embodiment, laser control system **550** may continue to track and service an ember until a discontinuity in that ember's thermal signature is observed by laser control system **550**. Once that discontinuity has been observed, laser control system **550** determines the next ember, in terms of priority, and disengages from the depleted ember to engage the next ember. This process may be repeated until all embers that pose an ignition risk to static structure **100** have been serviced and depleted.

(39) An embodiment of this method or sequence is shown in FIGS. **11-13**. Referring to FIGS. **11-13**, a group of oncoming embers is shown. Fourth ember **516** includes fourth thermal signature **1102** and a fourth fuel level **1104**. In FIGS. **11-13**, fourth fuel level **1104** is shown schematically as a gasoline gauge, which represents a quantity of unburned hydrocarbons, most likely wood fuel, associated with fourth ember **516**. Third ember **514** is also shown, having a third thermal signature **1106** and its associated third fuel level **1108**. Sixth ember **520** (not shown in FIG. **5**) is also shown, having sixth thermal signature **1110** and also having a sixth fuel level **1112** that represents a quantity of unburned hydrocarbons contained within sixth ember **520**. A number of other embers are also shown. Seventh ember **1114**, eighth ember **1116**, ninth ember **1118** and tenth ember **1120** are also shown. These embers have already depleted their corresponding fuel levels and do not

produce a significant thermal signature.

(40) Given this collection of incoming embers, embodiments of the present invention may operate to prioritize and service those embers that pose a fire risk to static structure **100**. Referring to FIGS. **4** and **6**, laser control system **550** may track these incoming embers, as shown in step **402**.

According to one embodiment, laser control system **550** may select a particular ember for initial service, as shown in step **404**. In this case, the ember with the highest thermal signature is selected for initial service. Recall that other embodiments may optionally include the velocity and trajectory of the ember in the ember selection process.

(41) In the embodiment shown in FIG. **11**, fourth ember **516** has the “brightest” thermal signature **1102**, which causes laser control system **550** to select the fourth ember **516** for initial service. Laser control system **550** maneuvers and guides laser emitter **502** towards fourth thermal signature **1102** and services fourth ember **516**. The optical adjustment and correction steps shown in FIGS. **7-9** may optionally be utilized to enhance the accuracy of laser emitter **502**.

(42) After fourth ember **516** has been sufficiently depleted, laser control system **550** may select the next target ember for service. This situation is shown in FIG. **12**. In the embodiment shown in FIG. **12**, third ember **514** has the next highest or “brightest” thermal signature **1106**. This causes laser control system **550** to select the third ember **514** for subsequent service. Laser control system **550** shifts the aim of laser emitter **502** from fourth ember **516** to third ember **514**, and commands laser emitter **502** to direct laser energy towards third ember **514**. Again, the optical adjustment and correction steps shown in FIGS. **7-9** may optionally be utilized to enhance the accuracy of laser emitter **502** as it services third ember **514**. The laser energy accelerates the dissipation of the third ember's **514** corresponding fuel level **1108**. Eventually, third fuel level **1108** becomes depleted and third ember **514** experiences a significant drop in thermal signature.

(43) Meanwhile, sixth ember **520** which initially had the third brightest thermal signature **1110** (see FIG. **11**) after the thermal signatures the fourth and third embers, has burn out on its own without any intervention. Note that in FIG. **12**, the sixth fuel level **1112** has become depleted while the third ember **514** was being serviced. FIG. **13** is an embodiment of the situation after the third ember **514** has been serviced. In this embodiment, all of the incoming embers have been neutralized and depleted of fuel. Their lack of fuel prevents those embers from supporting further combustion or fire. This situation can also be observed in step **408** (See FIG. **4**) where the ember fuel has been depleted and the system returns to step **402**, waiting to track any new incoming embers. FIGS. **1-13** show an embodiment of the system engaging an incoming group of embers. After this engagement has concluded, the system may wait at step **402** (See FIG. **4**) for any new incoming embers and repeat the process again and again until the threat from incoming embers has been fully mitigated. Eventually, the desired goal would be to deplete the fuel of all incoming embers that pose a threat so that none of the incoming embers presents a significant risk of igniting static structure **100**, and the overall threat from embers has been mitigated by using the principles and features disclosed with the various embodiments.

(44) The processes and methods of the embodiments described in this detailed description and shown in the figures can be implemented using any kind of computing system having one or more central processing units (CPUs) and/or graphics processing units (GPUs). The processes and methods of the embodiments could also be implemented using special purpose circuitry such as an application specific integrated circuit (ASIC). The processes and methods of the embodiments may also be implemented on computing systems including read only memory (ROM) and/or random access memory (RAM), which may be connected to one or more processing units. Examples of computing systems and devices include, but are not limited to: servers, cellular phones, smart phones, tablet computers, notebook computers, e-book readers, laptop or desktop computers, all-in-one computers, as well as various kinds of digital media players.

(45) The processes and methods of the embodiments can be stored as instructions and/or data on non-transitory computer-readable media. The non-transitory computer readable medium may

include any suitable computer readable medium, such as a memory, such as RAM, ROM, flash memory, or any other type of memory known in the art. In some embodiments, the non-transitory computer readable medium may include, for example, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of such devices. More specific examples of the non-transitory computer readable medium may include a portable computer diskette, a floppy disk, a hard disk, magnetic disks or tapes, a read-only memory (ROM), a random access memory (RAM), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), an erasable programmable read-only memory (EPROM or Flash memory), electrically erasable programmable read-only memories (EEPROM), a digital versatile disk (DVD and DVD-ROM), a memory stick, other kinds of solid state drives, and any suitable combination of these exemplary media. A non-transitory computer readable medium, as used herein, is not to be construed as being transitory signals, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

(46) Instructions stored on the non-transitory computer readable medium for carrying out operations of the present invention may be instruction-set-architecture (ISA) instructions, assembler instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, configuration data for integrated circuitry, state-setting data, or source code or object code written in any of one or more programming languages, including an object oriented programming language such as Smalltalk, C++, or suitable language, and procedural programming languages, such as the “C” programming language or similar programming languages.

(47) Aspects of the present disclosure are described in association with figures illustrating flowcharts and/or block diagrams of methods, apparatus (systems), and computing products. It will be understood that each block of the flowcharts and/or block diagrams can be implemented by computer readable instructions. The flowcharts and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of various disclosed embodiments. Accordingly, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions. In some implementations, the functions set forth in the figures and claims may occur in an alternative order than listed and/or illustrated.

(48) The embodiments may utilize any kind of network for communication between separate computing systems. A network can comprise any combination of local area networks (LANs) and/or wide area networks (WANs), using both wired and wireless communication systems. A network may use various known communications technologies and/or protocols. Communication technologies can include, but are not limited to: Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), mobile broadband (such as CDMA, and LTE), digital subscriber line (DSL), cable internet access, satellite broadband, wireless ISP, fiber optic internet, as well as other wired and wireless technologies. Networking protocols used on a network may include transmission control protocol/Internet protocol (TCP/IP), multiprotocol label switching (MPLS), User Datagram Protocol (UDP), hypertext transport protocol (HTTP), hypertext transport protocol secure (HTTPS) and file transfer protocol (FTP) as well as other protocols.

(49) Data exchanged over a network may be represented using technologies and/or formats including hypertext markup language (HTML), extensible markup language (XML), Atom, JavaScript Object Notation (JSON), YAML, as well as other data exchange formats. In addition, information transferred over a network can be encrypted using conventional encryption technologies such as secure sockets layer (SSL), transport layer security (TLS), and Internet Protocol security (Ipsec).

(50) While various embodiments of the disclosure have been described, the description is intended to be exemplary, rather than limiting and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the

disclosure. Accordingly, the disclosure is not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

Claims

1. A static structure including a laser system comprising: at least two vertically disposed side walls, the side walls being angled with respect to one another; a roof supported by the two side walls and generally disposed above the two side walls; a laser system including a laser turret; the laser turret being disposed on the roof of the static structure; the laser system including a sensor array and a laser emitter configured to dispense laser energy; the laser turret also including a control system configured to move the laser emitter; a laser control system receiving information from the sensor array and sending signals to the control system to move the laser emitter; wherein the laser control system selects an incoming ember based on a thermal signature associated with the incoming ember, and wherein the laser control system issues commands to the control system to direct the laser emitter towards the incoming ember, and wherein the laser emitter directs laser energy at the incoming ember; and wherein the laser control system uses a camera sensor of the sensor array to correct the direction of the laser emitter to follow the incoming ember while the incoming ember moves, after the incoming ember has been initially acquired by the thermal signature associated with the incoming ember.
2. The static structure according to claim 1, wherein the laser control system tracks multiple incoming embers and calculates a velocity and trajectory of each of the incoming embers.
3. The static structure according to claim 1, wherein the laser control system determines that a fuel level associated with the incoming ember has been depleted by sensing a discontinuity in the thermal signature of the incoming ember.
4. The static structure according to claim 1, wherein the laser control system determines that a fuel level associated with the incoming ember has been depleted by sensing a rapid decline in the thermal signature of the incoming ember.
5. The static structure according to claim 1, wherein the laser turret includes a rotating motor that rotates the laser turret with respect to the static structure.
6. The static structure according to claim 1, wherein sensor array includes a camera sensor and a reflecting sensor.
7. A static structure including a fire mitigation system comprising: at least two vertically disposed side walls, the side walls being angled with respect to one another; a roof supported by the two side walls and generally disposed above the two side walls; a laser system including a laser turret; the laser turret being disposed on the static structure; the laser system including a sensor array and a laser emitter configured to dispense laser energy; the laser turret also including a control system configured to move the laser emitter; a laser control system receiving information from the sensor array and sending signals to the control system to move the laser emitter; wherein the laser control system tracks multiple incoming embers and calculates a velocity and trajectory of each of the incoming embers, the laser control system selecting at least one priority ember likely to contact the static structure for initial service, and wherein the laser control system directs laser energy at the priority ember.
8. The static structure according to claim 7, wherein the laser control system continues to direct laser energy at the priority ember until a fuel level associated with the priority ember has been depleted.
9. The fire protection system of claim 8, wherein the fire protection system controller further comprises a light sensor to locate embers.
10. The fire protection system of claim 8, wherein the switch is adapted to separately activate selected laser light emitters.

11. The fire protection system of claim 10, wherein a first laser light emitter is aimed at and tracks the first ember.
 12. The fire protection system of claim 11, wherein a second laser light emitter is aimed at and tracks the first ember.
 13. The fire protection system of claim 8, wherein the controller is adapted to locate and track with emitted laser light from a first laser light emitter an ember presenting greatest risk to the building to neutralize the ember.
 14. The fire protection system of claim 13, wherein the controller is further adapted to track with emitted laser light from a second laser light emitter an ember presenting greatest risk to the building to neutralize the ember.
 15. A method for neutralizing embers that may burn a building, the method comprising: scanning a region adjacent to the building to locate a first ember and a second ember both having a risk of burning the building; selecting an ember having the greater risk of burning by using a velocity and trajectory of the ember; tracking the ember selected to have the greater risk of burning the building; and energizing a laser light emitter to emit laser light; directing emitted laser light on the ember for a period sufficient to neutralize the ember.
 16. The method of claim 15, further comprising a light sensor used to track incoming embers.
 17. The method of claim 16, further comprising: a turret configured to direct emitted laser light on the selected ember.
 18. The method of claim 16, further comprising energizing the laser light emitter continuously until all embers are neutralized.
 19. The method of claim 17, further comprising energizing the laser light emitter only when the laser is directed at the selected ember, so that the laser is energized discontinuously and is turned off between embers.
 20. The method of claim 15, wherein a camera sensor is used to aim the laser light emitter.
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